

ABSTRACT

This paper discussed the design construction, verification testing and application of passive RFID smart label for healthcare applications. It addresses specific key design characteristics and qualification testing methods and results to ensure the label prevents the potential spread of blood borne pathogens. This paper assumes the reader is well versed or otherwise not particularly interested in specifics of passive RFID technology as that is only in the context of the design construction and testing.

Table of Contents

ABSTRACT	1
Table of Figures	1
INTRODUCTION	2
BACKGROUND.....	2
PASSIVE RFID LABEL DESIGN	2
PASSIVE RFID LABEL QUALIFICATION TESTING.....	7
CONCLUSIONS AND RECOMMENTATIONS	9

Table of Figures

Figure 1. Omni ID® IQ 350 Core Product Specs	3
Figure 2. Omni ID® IQ350 Construction	3
Figure 3. Molecular Structure of PET Chemical Formula (C ₁₀ H ₈ O ₄) _n . Source: https://omnexus.specialchem.com	4
Figure 4. Closed- and open-cell foam structures.....	5
Figure 5. Close up of the exposed side of the IQ 350 closed cell foam	5
Figure 6. Close up vie of black closed cell foam used with a Zebra BLE beacon for healthcare ...	6
Figure 7. Healthcare Smart Label Example	6
Figure 8. IQ 350 qualification thermal test and result summary.....	7
Figure 10. Pre and thermal frequency shift data after 700 hrs soak at -20C mounted to a flat metal plate	8
Figure 11. Post thermal frequency shift data after 700 hrs soak at -20C mounted to a flat metal plate	8
Figure 9. IQ 350 IP 68 qualification test and result summary	9
Figure 10. Frequency shift data before being submerged in water for1 hr at a 1 m depth mounted to a flat metal plate.....	9
Figure 11. Frequency shift data after being submerged in water for1 hr at a 1 m depth mounted to a flat metal plate.....	9

INTRODUCTION

In today's healthcare environment the delivery of care depends on a multitude of variables orchestrated by skilled and knowledgeable clinicians. Having the right, well-maintained medical devices available when needed is assumed but can't be taken for granted. Managing medical devices throughout a facility may at first glance seem a mundane task, but it is critical to the safe delivery of care. Biomedical technicians are typically responsible for the tracking and maintenance of medical devices through one or more facilities. Knowing the physical location of devices is paramount to performing routine maintenance, repair, calibration and ensuring devices are ready and available when needed. The low per unit cost and lack of a battery make passive RFID technology ideal for asset management and tracking as done by biomedical technicians. Furthermore, passive RFID technology easily extends into complimentary workflows such as cleaning and disposal, staff and patient tracking, or other high value processes to help improve hospital operations overall. Passive RFID technology's low transmit power ensures smart labels can be mounted directly to sensitive medical devices without concern for electromagnetic interference.ⁱ This paper reviews a passive RFID based smart label design for asset tracking of medical equipment. It reviews the design construction and qualification testing of the label to address a specific concern regarding blood contamination and the potential spread of bloodborne pathogens.

BACKGROUND

According to the Occupational Safety and Health Administration Bloodborne pathogens are infectious microorganisms in human blood that can cause disease in humans.ⁱⁱ Hospitals and care facilities are required put processes in place to handle blood appropriately to minimize or even eliminate the exposure to bloodborne pathogens to ensure the safety of patients, employees, visitors etc. in accordance with OSHA's Bloodborne Pathogens Standard (29 CFR 1910.1030)ⁱⁱⁱ including cleaning and decontamination of equipment.^{iv} When medical equipment comes in contact with blood it needs to be cleaned by removing the blood and then being disinfected. This is also the case for the passive RFID or other labels and devices attached to the device. Designing a passive RFID smart label that will work both on metal and non-metal surfaces, withstand routine cleaning and disinfecting, and that will not absorb blood (or it's pathogens) is a challenge overcome with elegant design construction and thorough qualification testing.

PASSIVE RFID LABEL DESIGN

This is a brief narrative on the Omni IQ 350 label.

Omni ID® IQ 350 is part of its broad portfolio of printable, metal passive RFID labels. The IQ 350 is a small, low profile label that works both on and off metal. It has an excellent size-to-performance-ratio and has been qualified to work with leading thermal RFID printers (such as those made by Zebra). Below is a high-level specification for the IQ 350 passive RFID label.

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	IQ350
Read Range	3.5 m On Metal/ 2m off metal
Operational band	Regional
IC	Monza R6-P Only
Asset Type	On metal
Material	Foam Core
Ingress protection	IP68
Temperature	-40 to 85 °C
Size	50 mm x 12.5 mm x 1.35 mm

Figure 1. Omni ID® IQ 350 Core Product Specs

passive RFID smart label construction and material choices are the basis of a design that will work in healthcare and address the risk of potentially carrying or spreading infections. The following figures show both the material stack-up and construction of the IQ 350.

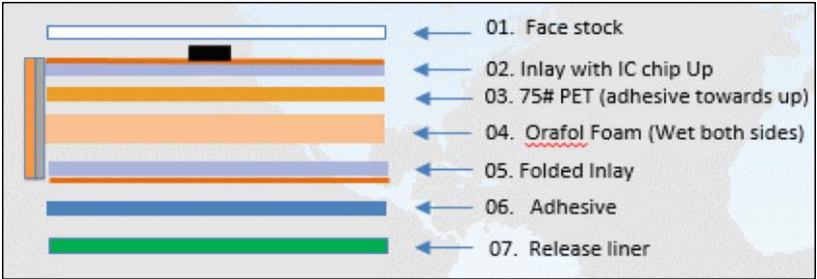


Figure 2. Omni ID® IQ350 Construction

The actual material selection is critical. To be used as a smart label in healthcare the materials must not only possess excellent RF performance characteristics but also hold up to routine

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cleaning with strong disinfectants. The face stock, adhesives and other materials must not be absorbent, and the inks used in the printing of the smart label must be permanent of high resolution and not react with disinfectants. Of course the competitive global market demands smart labels be easy to apply and cost effective.

The IQ 350 uses a polyethylene terephthalate (PET) material for the face stock and substrate material. These are layers 01 and 03 in the above figure. PET is part a thermoplastic polymer. It's part of the polyester family of polymers, has excellent resistance to alcohols and solvents, and is dimensionally stable. PET is found in common store bough water bottles, and it is one of the most recycled thermoplastics. Below is a figure of the molecular structure of a PET formula.

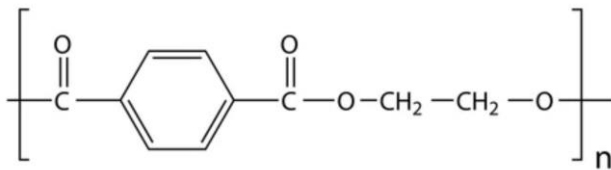


Figure 3. Molecular Structure of PET Chemical Formula $(\text{C}_{10}\text{H}_8\text{O}_4)_n$. Source: <https://omnexus.specialchem.com>

The IQ 350 has been optimized for RF performance when applied to metal and non-metal surfaces. This essentially requires the label to provide its own ground plane. To effectively realize an antenna structure with the gain and performance required there must be an electrical separation between the ground plane and the antenna structure. This is achieved by inserting a thin layer of closed cell foam between the inlay (includes the antenna structure) and the adhesive mounting surface. This is layer 04 in the preceding figure. The permittivity of the foam and thus the thickness of the foam must be tightly controlled to ensure a proper RF match for the antenna relative to the integral ground plane. It's also absolutely critical in this case that a closed cell foam be used. Open cell foams have entirely open spaces throughout the physical structure which can absorb and fill with liquids such as blood. Think of open-cell foams almost like a "3D wire mesh structure consisting mostly of open space." Closed-cell foam structures are typically formed when a material is subjected to a gas under high pressure. Small closed cells form one next to or near one another. Each cell (or "bubble") is a completely enclosed unit typically with gas trapped inside. The closed-cell foam structure is thus made up of millions of these small cells side-by-side but each sealed off from one another and spaces between the cells are filled with the material (there are no "open spaces"). There will be a certain rough appearance to the texture of the edge closed-cell foam, but the closed cell is in fact a non-absorbant liquid barrier. The following diagram shows a graphical comparison of open and closed cell structures.

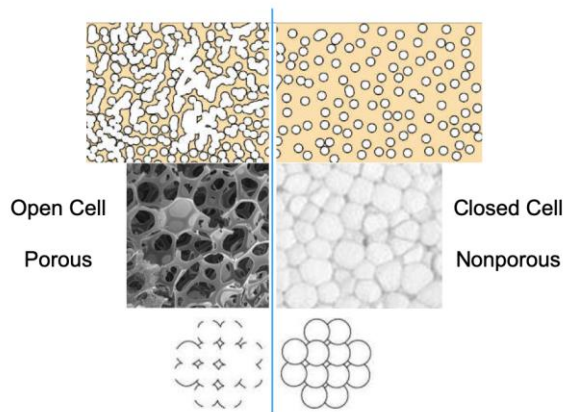


Figure 4. Closed- and open-cell foam structures

In the healthcare case the choice of a closed-cell polyethylene (PE) foam is critical. PE foam has the following material characteristics: high sheer strength, high temperature resistance, water resistance, and resistance to solvents. Closed cell PE foam is also naturally antimicrobial inhibiting the growth of bacteria and thus is ideal for use in smart labels for healthcare applications. The following figure is a close up of the closed-cell edge of an IQ 350 smart label.



Figure 5. Close up of the exposed side of the IQ 350 closed cell foam

The following is a separate example of an active RFID (Bluetooth® low energy) beacon that was designed specifically to withstand the rigors of regular cleaning in healthcare (Zebra's MPACT-MB1000-01-WR). Note that the closed cell foam in this example is black rather than white exhibits the same roughly looking exposed edge up close.

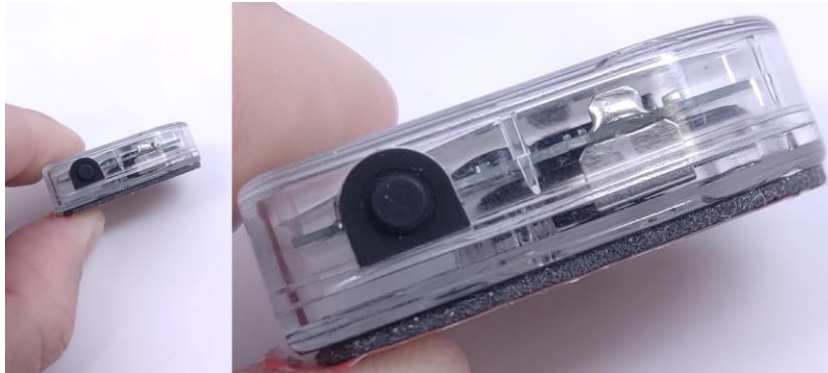


Figure 6. Close up vie of black closed cell foam used with a Zebra BLE beacon for healthcare

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The following is an example of a custom smart label design for GE Healthcare built upon the Omni ID® IQ 350. In this case it is an asset tracking label that includes human readable text, and asset identification number, a 2D barcode representing the asset identification number, and the internal RFID inlay encoded with the same asset identification number. The printing and encoding is done in a high-volume manufacturing process, and the asset identification numbers are in succession and unique for each label.



Figure 7. Healthcare Smart Label Example

PASSIVE RFID LABEL QUALIFICATION TESTING

Understanding the specific and perhaps unique healthcare requirements is only part of the passive RFID smart label development process. As discussed herein the construction of the label and the selection of materials is critical to an effective design that not only meets the requirements but can be produced in high-volume production at a reasonable cost. However, once the design is complete it must be tested to verify it meets the requirements. Certainly the label must meet various quality, cost, performance and other requirement; however in this case we are referring specifically to the requirement that the label must not absorb blood and must prevent the potential spread blood borne pathogens.

Qualification testing of the design is broken down in to two separate series of test. The first is thermal testing. Thermal testing is a means of testing the design construction. Baseline performance is measured, the labels are exposed to various time and temperature extremes and then evaluated for appearance (delamination) and performance changes (peak frequency shift). A shift in a label's RF center frequency is the best indicator of a performance change as a result of a qualification test. Reasonable center frequency shift criteria based on past experience were established for each test performed. The actual thermal qualification test sample quantities, conditions and results are summarized in the below table. Note also that the labels samples were attached to either a flat metal plate of metal pipe before being tested as they would when being used in the field as smart labels, and post test samples were evaluated for an delamination of label layers, bubble formation or dramatic color changes.

Product	Test Condition	Thermal Test	Qty	RF performance test result		Appearance check
				Peak frequency shift	Result	
IQ350EU R6P	On flat metal plate (Aluminium material)	700hrs thermal soak 85°C	5	shift up 2MHz	100% pass	No delamination, No bubbles
		700hrs 261 cycles thermal cycle -20~85°C	5	shift up 2MHz	100% pass	No delamination, No bubbles
		700hrs UV test	5	shift up 4MHz	100% pass	The color have no obvious changed
		700hrs thermal soak -20°C	5	shift up 4MHz	100% pass	No delamination, No bubbles
	On metal pipe (Steel material) with 115mm diameter	700hrs thermal soak 85°C	5	shift up 6MHz	100% pass	No delamination, No bubbles
		700hrs 261 cycles thermal cycle -20C to 85°C	5	shift up 6MHz	100% pass	No delamination, No bubbles
		700hrs thermal soak -20°C	5	shift up 4MHz	100% pass	No delamination, No bubbles

Figure 8. IQ 350 qualification thermal test and result summary

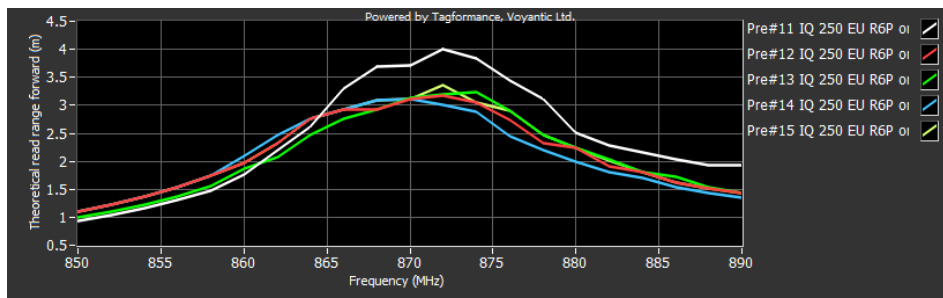


Figure 9. Pre and thermal frequency shift data after 700 hrs soak at -20C mounted to a flat metal plate

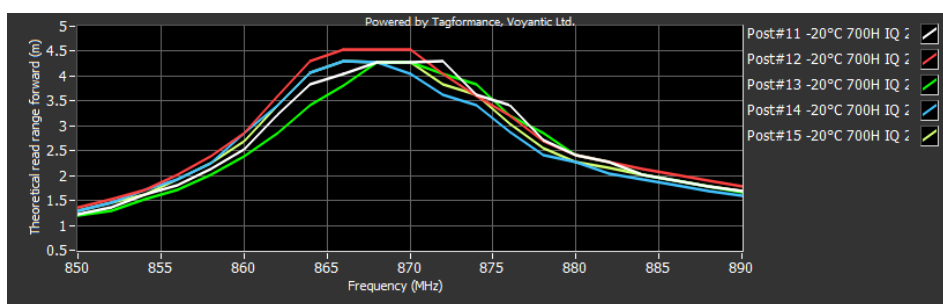


Figure 10. Post thermal frequency shift data after 700 hrs soak at -20C mounted to a flat metal plate

A critical element preventing the potential blood borne pathogens is the choice of appropriate synthetic PE and PET materials and use of closed-cell foam. From a design validation standpoint a test method for the absorption of liquids is an appropriate proxy for the absorption of blood. Stated differently, if the smart label does not absorb liquid water it should not absorb liquid blood. Ingress protection (IP) ratings are define the effectiveness against the intrusion of solid particulates (dust) and liquids. An IP68^v was determined to be an appropriate rating sufficiently preventing the absorption of blood: no dust ingress (the “6”, dust proof) and protection against continuous and complete water immersion (the “8”, waterproof).

The second qualification test to validate an IP 68 rating entails submersing the tag in a tank of water at a depth of one meter (1m) for a period of one hour (1hr). A label is first encoded/printed before baseline performance measurements. Next the label is submerged as described herein. The label is then removed from the water and the baseline tests are repeated. Lately the baseline and post immersion test results are compared. Pre and post-immersion performance changes are then compared against a pre-determined criteria. Again, a shift in a label’s RF center frequency is the best indicator of a performance change as a result of this qualification test. If the integrity of the closed-cell PE foam is compromised—i.e. the physical closed-cell structure fails and exchanges the cells’ own internal gas for liquid—then the electrical permittivity of the foam will change. The relative permittivity of the foam (it’s

dielectric constant) is a measure of the foam’s capacity to store electric charge. Changes in the relative permittivity of the foam (e.g. if it absorbs water) will in turn change the RF antenna match and substantially shift the RF center frequency.

Product	Test Condition	Test Item	Sample Quantity	RF performance post IP68 test		Appearance check
				Peak RF Freq. shift	Result	
IQ350	On flat metal plate (Aluminium material)	IP68 test	2	shift up <= 2MHz	100% pass	No obvious change

Figure 11. IQ 350 IP 68 qualification test and result summary

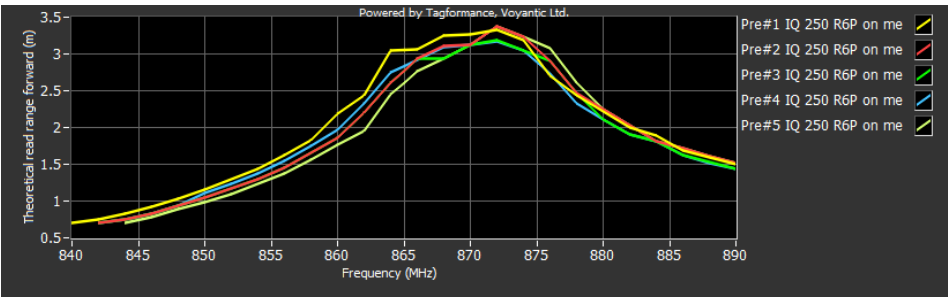


Figure 12. Frequency shift data before being submerged in water for 1 hr at a 1 m depth mounted to a flat metal plate

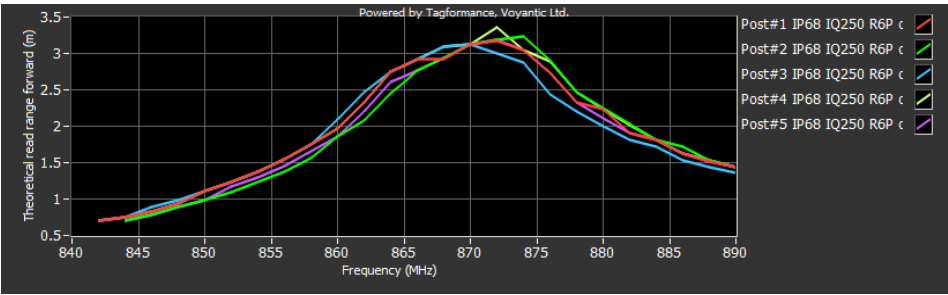


Figure 13. Frequency shift data after being submerged in water for 1 hr at a 1 m depth mounted to a flat metal plate

The qualification testing of the IQ 350 confirms that the product is sufficiently rugged and meets its IP 68 rating. The product will not absorb liquids like water or blood.

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CONCLUSIONS AND RECOMMENDATIONS

The IQ 350 passive RFID label is specifically designed and tested to survive as smart labels for healthcare; and the label’s closed cell, nonporous foam and other synthetic materials are non-porous and do not absorb liquids such as blood. If soaked with blood the labels should be cleaned

in accordance with all appropriate worksite processes/procedures^{vi}; and workers should follow all relevant safety precautions and controls such as OSHA's Bloodborne Pathogens Standard (29 CFR 1910.1030)^{vii}. These smart labels will survive exposure to blood. They should not absorb or retain liquid blood. Normal hospital procedures for cleaning assets exposed to blood should alleviate contamination with blood borne pathogens or other infections material. These smart labels should be disinfected regularly following a hospital's normal cleaning processes as would be done for other nonporous surfaces such as the medical device surfaces the label is adhered to. We would expect the normal cleaning process will adequately disinfect the surfaces of the label which is where the liquid blood would exist (blood should not be absorbed into the PE foam).

If the smart label get's damaged—ripped, torn, gouged or otherwise physically compromised—it is always best to remove the label, clean the surface appropriately and apply a new label. However, if the physical label is not damaged it can be cleaned with strong disinfectants and used with confidence knowing that it's been specifically designed and tested to be non-absorbent. These smart labels will not absorb blood, and with proper cleaning they will not spread blood borne pathogens.

ⁱ Before applying any sort of passive or active tracking device to medical equipment one should always review the medical device manufacturer's EMI limits to ensure compliance with a safe separation distance for that medical device.

ⁱⁱ <https://www.osha.gov/SLTC/bloodbornepathogens/>

ⁱⁱⁱ <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1030>

^{iv} https://www.dir.ca.gov/dosh/dosh_publications/BBPBest1.pdf

^v Ingress protection ratings are defined in the international standard EN 60529 (British BS EN 605:1992, European IEC 6059:1989). <http://www.enclosurecompany.com/ip-ratings-explained.php>

^{vi} Center for Disease Control and Prevention; Guideline for Disinfection and Sterilization in Healthcare Facilities, updated May 2019, <https://www.cdc.gov/infectioncontrol/guidelines/disinfection/>

^{vii} United States Department of Labor, Occupational Safety and Health Administration; <https://www.osha.gov/SLTC/bloodbornepathogens/>

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Stains and stain removal? Aesthetics
Add overlay
Change color of foam
Encapsulated tag